



(Don't Write anything in this Area)

Section A

Q.1) Write notes on the following in 150 words

10*5=50

1.a) Hardy-Weinberg Law

The law was proposed by Hardy & Weinberg which forms a central idea in population genetics.

The law states that the genetic & genotypic frequency of a given population shall remain constant as long as the evolutionary forces do not act upon them.

However, it assumes following conditions

- 1) Population is a closed system
- 2) There is no gene flow from external
- 3) Every individual has equal chance of mating with every other i.e. no segregation
- 4) Not exposed to evolutionary forces.

Mathematical Interpretation

the law can be mathematically

represented as,

let p = frequency of one allele of a gene

q = frequency of another allele of same gene

$\therefore \boxed{p + q = 1}$ i.e. Genetic frequency is constant

then, the genotypic frequency will be

$$\boxed{p^2 + q^2 + 2pq = 1}$$

$p^2 \rightarrow$ homozygous dominant condition

$q^2 \rightarrow$ homozygous recessive "

$pq \rightarrow$ heterozygous condition.

Relevance

1) Weinberg differential method - used in determining proportion of dizygotic & monozygotic in twin studies

2) Law is crucial in evolutionary studies & helps determine the impact of evolutionary forces

3) Population genetics - helps in determining the variations of human body at both intra & inter population level.

The law is also used in

Applied anthropology (Medical anthropology) showing its practicality